Project 2

1. Introduction

This report details the development of a logistic regression model to predict species classification for an unclassified dataset of Microtus skull measurements. The dataset for Project 2 encompasses three variables across two known species, M. subterraneus and M. multiplex, as well as an unclassified set of the Microtus genus. **We looked at the summary of the total dataset without any editing**.

The known dataset **without any cleaning** includes 44 observations for M. multiplex and 46 for M. subterraneus, with measurements of skull length, height above bullae, and width across the rostrum. An additional 199 observations remain unclassified. The goal of this study is to apply the logistic regression model based on the 90 known observations to predict the classification of the 199 unclassified observations.

2. Understanding the Dataset

We combined the datasets of M. subterraneus and M. multiplex, along with the 'Unknown' category, into a single comprehensive dataset. This inclusive approach was crucial for gaining a more complete understanding of the dataset.

To effectively implement the logistic regression model for species classification, we merged the datasets of M. subterraneus and M. multiplex into a single dataset, while maintaining the 'Unknown' category as a separate entity.

By combining the known species data, we were able to establish a robust baseline for the model, facilitating more accurate predictions when applied to the 199 unclassified observations.

In summary, the overall dataset of 289 entries has been categorized based on the species. 44 of these are well-defined under 'Multiplex' and 46 are under 'Subterraneus', while a larger portion, 199 entries, remain under 'Unknown', indicating areas where characteristics are not clearly defined.

Species	Multiplex	Subterraneus	Unknown
Counts	44	46	199

Dataset Information	Total
Combined (Multiplex and Subterraneus)	90
Unknown	199
Total Dataset	289

Table 1: Original Dataset Information

Total Dataset (289 Entries): Encompasses ‘Multiplex' 'Subterraneus' and 'Unknown'.

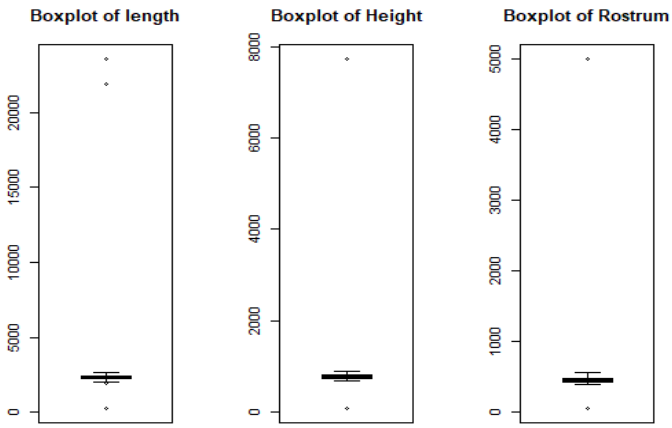
Total Dataset Combined (Multiplex and Subterraneus) (90 Entries): Focuses on entries labeled 'Multiplex' or 'Subterraneus', representing a specific subset of the total dataset.

Unknown Dataset (199 Entries): Consists of data that couldn't be classified as either 'Multiplex' or 'Subterraneus', forming a major portion of the dataset with ambiguous characteristics.

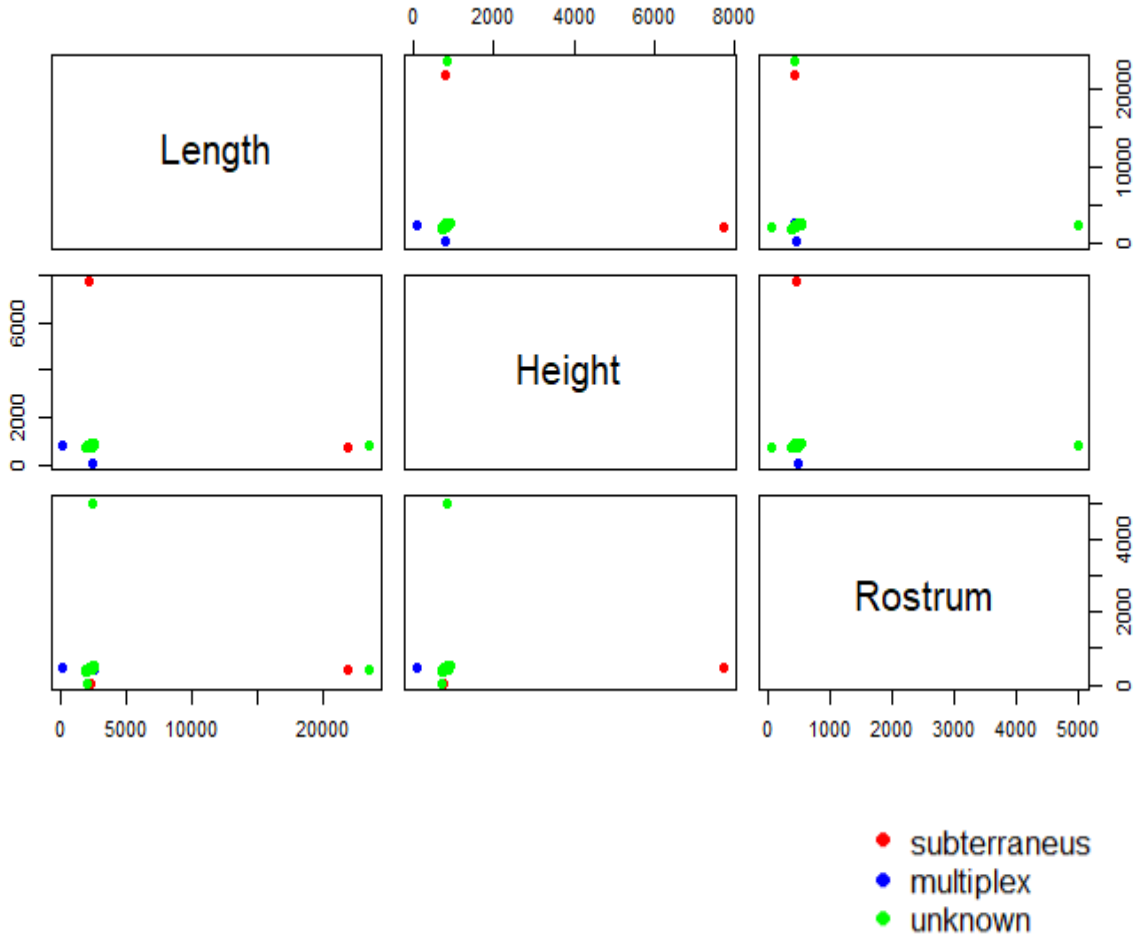
Species	Multiplex	Subterraneus	Unknown
Counts	44	46	199

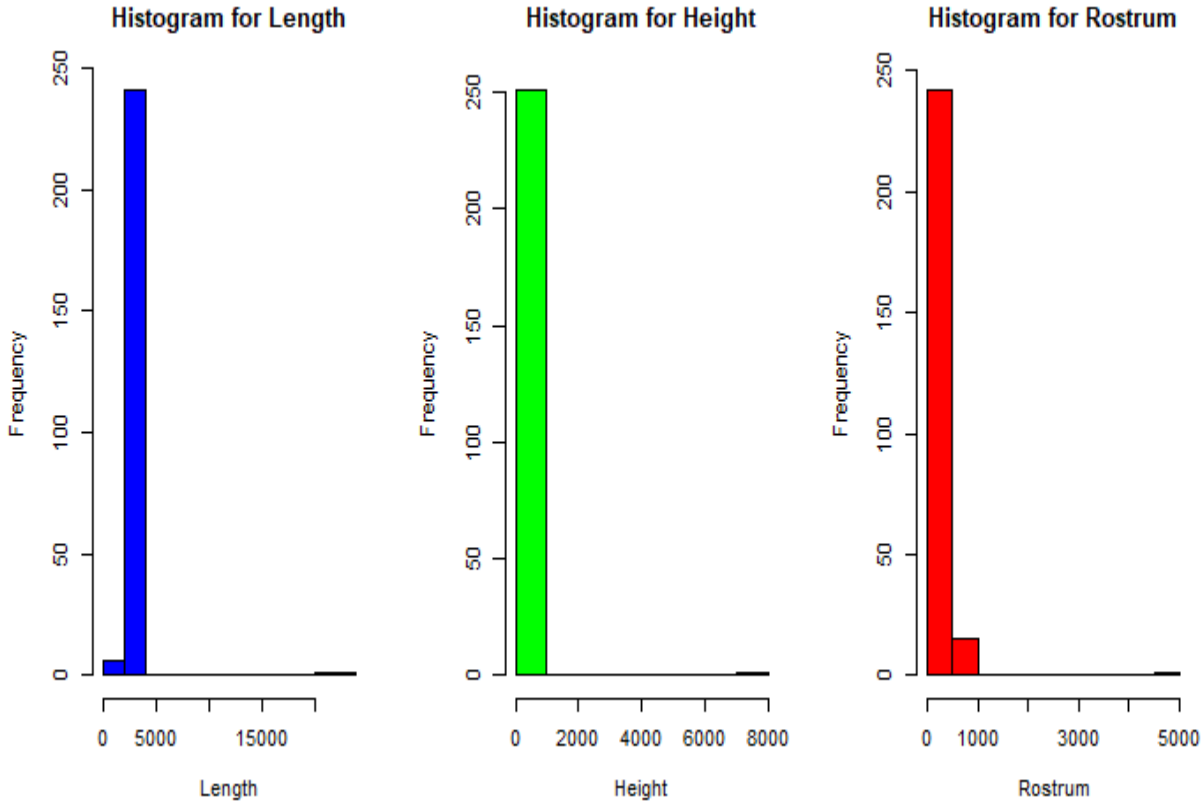
Variables	MIN.	1ST QU.	MEDIAN	MEAN	3RD QU.	MAX.	NA'S
LENGTH	234	2221	2312	2462	2390	23555	41
HEIGHT	84	755	790	815.3	820	7722	38
ROSTRUM	40	425	450.5	465.3	475	5000	32

Table 1: Summary Statistics of Total Dataset



Scatterplot Matrix of Total Dataset





Upon examining the boxplot and summary statistics of the total dataset, we identified several potential outliers. However, a closer investigation into the dataset trends suggested that these outliers were likely due to data entry errors. As a result, we have made appropriate adjustments to account for these discrepancies.

This recalibration allows for a more accurate representation of the data, ensuring that our predictive analysis is based on reliable and pertinent information.

2.1 Data Cleaning Record

During the preliminary data examination, several issues were identified and addressed as follows:

Duplicates

M. multiplex Species Corrections

- **Vole Skull Index 29:** Identified as a duplicate entry in the dataset. Action taken: Removed the duplicate record to maintain the integrity of the dataset.
- **Vole Skull Index 8 (Height):** The recorded height was suspected to be a typographical error. Action taken: Corrected the height value from 84 to 845.
- **Vole Skull Index 39 (Length):** The recorded length appeared to be inconsistent with expected measurements. Action taken: Amended the length value from 234 to 2345.
- **Vole Skull Index:** One blank cell removed in between Vole skull Index 32 and 33.

M. subterraneus Species Corrections

- **Vole Skull Index 2 (Rostrum Width):** Detected a potential typo in the rostrum width measurement. Action taken: Altered the rostrum width from 42 to 425.
- **Vole Skull Index 13 (Length):** The length measurement was identified as a clear typographical error. Action taken: Adjusted the length value from 21899 to 2189.
- **Vole Skull Index 33 (Height):** The height measurement was notably erroneous. Action taken: Modified the height from 7722 to 772.
- **Vole Skull Index:** One blank cell removed in between Vole skull Index 12 and 13.

Unclassified Species Corrections (Unknown)

- **Vole Skull Index 7 (Rostrum Width):** The rostrum width was evidently incorrect. Action taken: Updated the rostrum width from 5000 to 500.
- **Vole Skull Index 10 (Rostrum Width):** Another instance of a rostrum width measurement being incorrect. Action taken: Changed the rostrum width from 40 to 405.
- **Vole Skull Index 45 (Length):** Detected an apparent typo in one of the measurements. Action taken: Changed the value from 23555 to 2355.
- **Blank Cell Correction:** A blank cell was found between records 20 and 21. Action taken: Removed the blank cell to ensure dataset continuity.

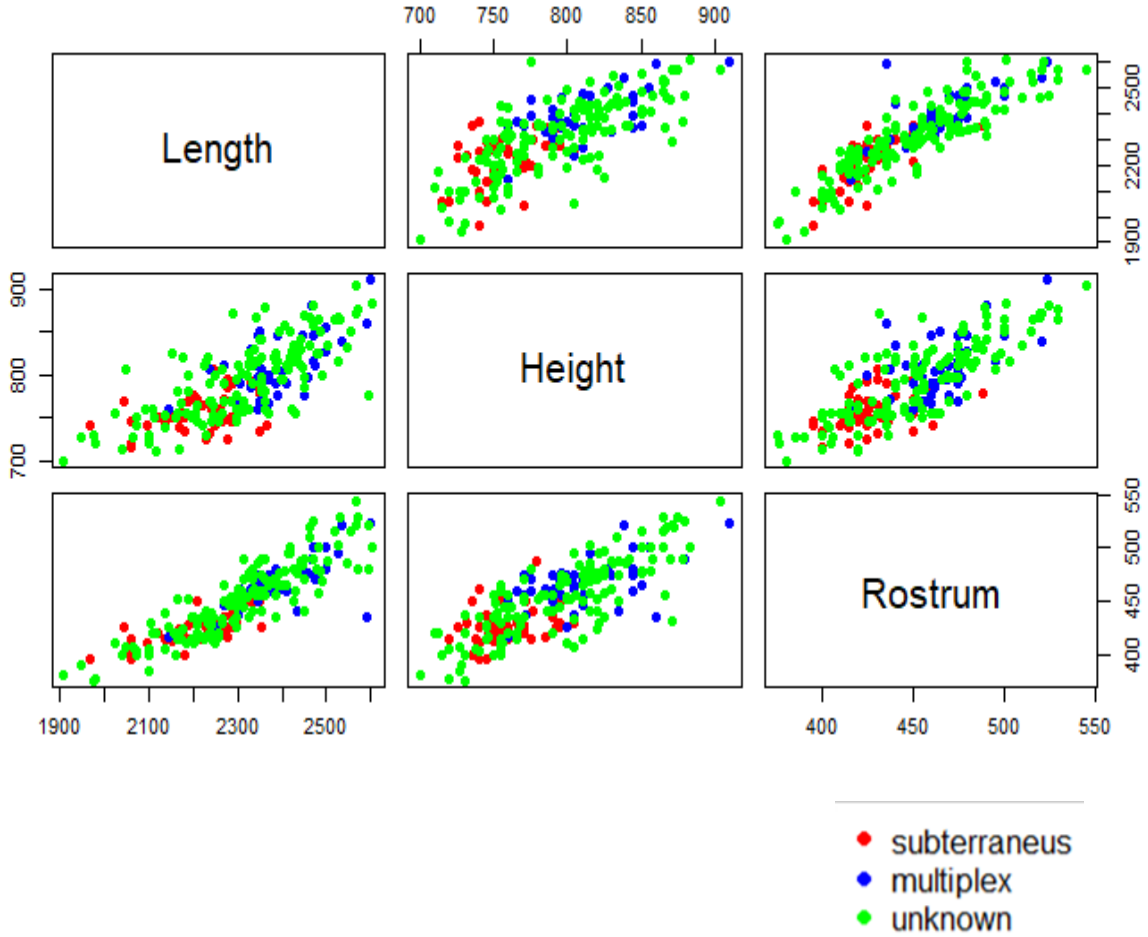
Each corrective action taken was after a thorough review of the data patterns for the species involved. The integrity of the data was maintained throughout, with the objective of creating the most accurate data set possible for the subsequent analysis.

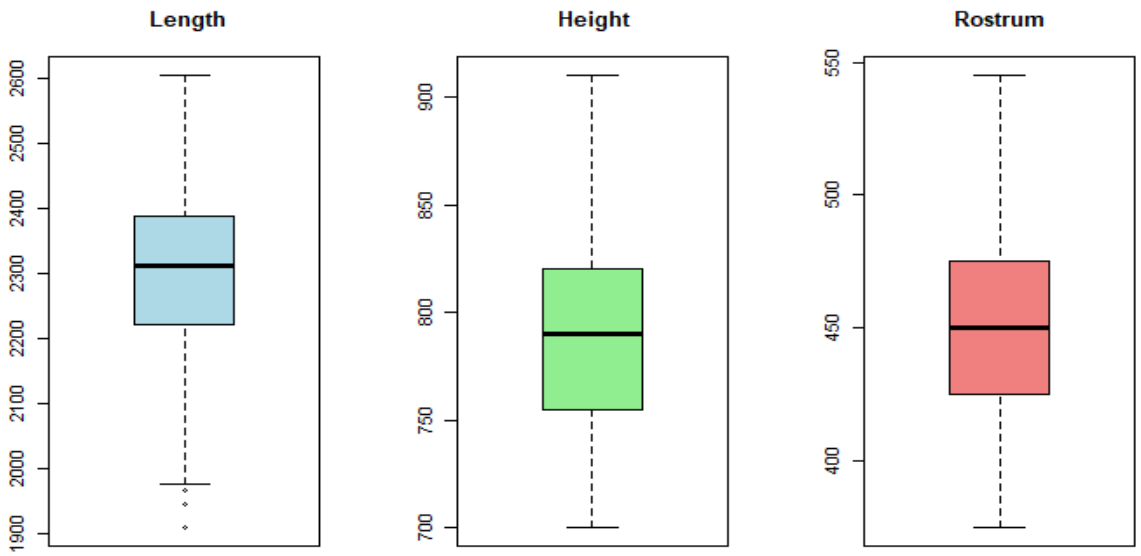
After cleaning the total dataset, we have the summary statistics in the table below:

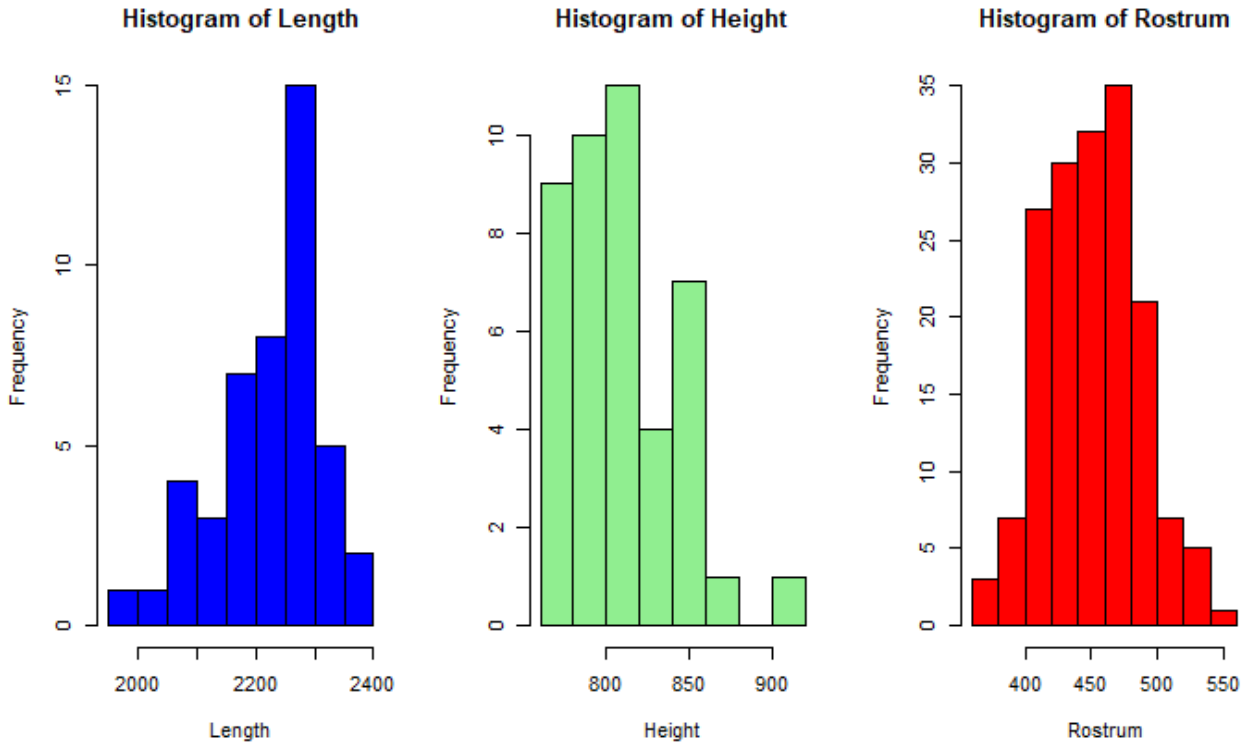
Species	Multiplex	Subterraneus	Unknown
Counts	43	46	199

Variables	Min	1st Qu	Median	Mean	3rd Qu	Max	NA's
Length	1908	2221	2312	2306	2388	2605	40
Height	700	755	790	790.8	820	910	37
Rostrum	375	425	450	450.8	475	545	31

Table 2: Summary Statistics (After cleaning)


Length
Height
Rostrum
700
750
800
850
900
1900
2100
2300
2500
2200
400
450
500
550
subterraneus
multiplex
unknown


Length
Height
Rostrum
1900
2000
2100
2200
2300
2400
2500
2600
700
750
800
850
900
400
450
500
550



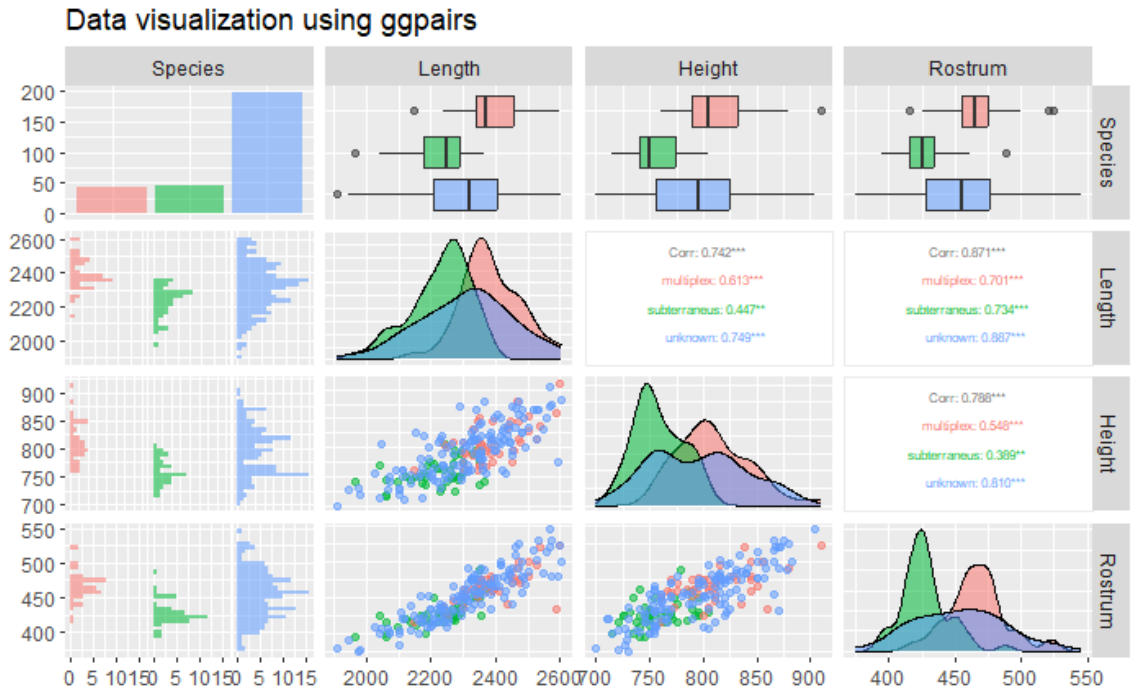
2.3 Correlation

Variables	Length	Height	Rostrum
Length	1	0.7384	0.8426
Height	0.7384	1	0.7296
Rostrum	0.8426	0.7296	1

- **Length and Height (Correlation: 0.7384):** There is a significant positive correlation, indicating that longer skulls tend to be taller.
- **Length and Rostrum (Correlation: 0.8426):** This shows a very strong positive correlation, suggesting that as skulls increase in length, they also tend to be wider across the rostrum.
- **Height and Rostrum (Correlation: 0.7296):** There is also a strong positive correlation here, indicating that taller skulls are generally wider across the rostrum.

The correlations suggest that the skull measurements are related; as one measurement increases, the others tend to increase as well. These relationships can be useful for predicting one measurement from the others and for understanding the overall skull structure of the species.

2.4 Data visualization Using GGpairs



3. Building Logistic Model

In this analysis, Leave-One-Out Cross-Validation (LOOCV) was employed to assess the model's performance and generalization ability. LOOCV is a well-established technique in statistical modeling and machine learning, involving systematically leaving out one data point at a time for testing while using the remaining data for model training (James, G., Witten, D., Hastie, T., & Tibshirani, R., 2013). This process is repeated for each data point, allowing for a comprehensive assessment of a model's performance and generalization ability.

Model 1
Summary Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.70439	-0.35385	0.04977	0.35551	2.35238

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	75.933029	17.224672	4.408	1.04e-05***
Length	-0.004264	0.007632	-0.559	0.57641
Height	-0.048627	0.017152	-2.835	0.00458 **
Rostrum	-0.062981	0.029761	-2.116	0.03432 *

Null deviance: 123.279 on 88 degrees of freedom
Residual deviance: 50.625 on 85 degrees of freedom
AIC: 58.625
Number of Fisher Scoring iterations: 6

Intercept: The intercept represents the estimated log-odds of the "multiplex" category when all other predictor variables are zero. In this case, the estimated log-odds of "multiplex" is 75.933029.

Length: The coefficient for "Length" is -0.004264. This means that for each one-unit increase in "Length," the log-odds of being in the "multiplex" category decrease by 0.004264 when all other variables are held constant. However, this coefficient is not statistically significant since the p-value is 0.57641, indicating that "Length" is not a significant predictor of the outcome.

Height: The coefficient for "Height" is -0.048627. This suggests that for each one-unit increase in "Height," the log-odds of being in the "multiplex" category decrease by 0.048627 when all other variables are held constant. This predictor is statistically significant because the p-value is 0.00458 (indicated by the ** symbol), suggesting that "Height" is a significant predictor of the outcome.

Rostrum: The coefficient for "Rostrum" is -0.062981. This means that for each one-unit increase in "Rostrum," the log-odds of being in the "multiplex" category decrease by 0.062981 when all other variables are held constant. This predictor is also statistically significant with a p-value of 0.03432 (* symbol).

Deviance Residuals: These residuals are a measure of how well the model fits the data. Deviance residuals indicate the difference between the observed and predicted values. Smaller residuals indicate a better fit.

Null Deviance: The null deviance represents the deviance when no predictors are included in the model (i.e., the intercept-only model). It serves as a baseline for comparing the fit of the model. In this case, the null deviance is 123.279.
Residual Deviance: The residual deviance represents the deviance of the model with the predictors included. A smaller residual deviance indicates a better fit of the model to the data. In the model, the residual deviance is 50.625, which is significantly smaller than the null deviance, suggesting that the model provides a better fit than the intercept-only model.
AIC (Akaike Information Criterion): AIC is a measure of the model's goodness of fit, taking into account the number of parameters. A lower AIC value indicates a better-fitting model. In the case, the AIC is 58.625.

Number of Fisher Scoring Iterations: This indicates the number of iterations required for the optimization algorithm to converge.

In summary, the logistic regression model suggests that "Height" and "Rostrum" are statistically significant predictors of the "multiplex" species category, while "Length" is not a significant predictor. The model has a lower residual deviance than the null deviance, indicating a good fit, and the AIC value suggests that this model is a reasonable choice for explaining the relationship between the predictor variables and the species categories.

Also '*' this mean significant at respective level *5% **1% ***0.01%.

Model 2

Summary Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.8437	-0.4545	0.0282	0.4307	1.7821

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	81.224353	18.242214	4.453	8.49e06***
Length	-0.017834	0.006192	-2.880	0.00398**
Height	-0.051243	0.016195	-3.164	0.00156**

Null deviance: 123.279 on 88 degrees of freedom
Residual deviance: 55.489 on 86 degrees of freedom
AIC: 61.489
Number of Fisher Scoring iterations: 6

Length: The coefficient for Length is -0.017834. It is statistically significant ($p < 0.05$), suggesting that Length has a significant effect on the log-odds of the response variable.
Height: The coefficient for Height is -0.051243. It is also statistically significant ($p < 0.05$), indicating that Height has a significant effect on the log-odds of the response variable.

Model Fit:
Null Deviance: 123.279 (baseline deviance)
Residual Deviance: 55.489 (model deviance)
AIC: 61.489 (model fit statistic)

Model 3

Summary Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.7154	-0.5261	0.0740	0.5078	2.5374

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	52.430564	11.057211	4.742	2.12e-06 ***
Length	-0.009630	0.005776	-1.667	0.09550
Rostrum	-0.067581	0.024008	-2.815	0.00488 **

Null deviance: 123.279 on 88 degrees of freedom
Residual deviance: 61.319 on 86 degrees of freedom
AIC: 67.319
Number of Fisher Scoring iterations: 6

Length: The coefficient for Length is -0.009630. It is not statistically significant at conventional significance levels (p-value > 0.05).

Rostrum: The coefficient for Rostrum is -0.067581. It is statistically significant ($p < 0.05$), suggesting that Rostrum has a significant effect on the log-odds of the response variable.

Model Fit:
Null Deviance: 123.279 (baseline deviance)
Residual Deviance: 61.319 (model deviance)
AIC: 67.319 (model fit statistic)

Model 4

Summary Deviance Residuals:

Min	**1Q**	**Median**	**3Q**	**Max**
-2.59371	-0.37456	0.07343	0.37216	2.48084

Coefficients:

	Estimate	**Std. Error**	**z value**	**Pr(>|z|)**
(Intercept)	72.97327	15.59640	4.679	2.88e-06 ***
Height	-0.05061	0.01663	-3.043	0.002340 **
Rostrum	-0.07496	0.02165	-3.462	0.000537 ***

Null deviance: 123.279 on 88 degrees of freedom
Residual deviance: 50.963 on 86 degrees of freedom
AIC: 56.963
Number of Fisher Scoring iterations: 6

Height: The coefficient for Height is -0.05061. It is statistically significant ($p < 0.01$), indicating that Height has a significant effect on the log-odds of the response variable.
Rostrum: The coefficient for Rostrum is -0.07496. It is also statistically significant ($p < 0.001$), suggesting that Rostrum has a significant effect on the log-odds of the response variable.

Model Fit:
Null Deviance: 123.279 (baseline deviance)
Residual Deviance: 50.963 (model deviance)
AIC: 56.963 (model fit statistic)

Model 5

Summary Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.84497	-0.65846	0.05418	0.63843	1.75442

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	55.964936	12.484745	4.483	7.37e-06 ***
Length	-0.024212	0.005401	-4.483	7.37e-06 ***

Null deviance: 123.28 on 88 degrees of freedom
Residual deviance: 69.39 on 87 degrees of freedom
AIC: 73.39
Number of Fisher Scoring iterations: 6

Length: The coefficient for Length is -0.024212. It is statistically significant ($p < 0.001$), indicating that Length has a significant effect on the log-odds of the response variable.

Model Fit:
Null Deviance: 123.28 (baseline deviance)
Residual Deviance: 69.39 (model deviance)
AIC: 73.39 (model fit statistic)

Model 6

Summary Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.4044	-0.5532	0.1268	0.5080	2.8260

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	42.16864	8.16529	5.164	2.41e-07***
Rostrum	-0.09456	0.01835	-5.164	2.55e-07***

Null deviance: 123.28 on 88 degrees of freedom
Residual deviance: 65.02 on 87 degrees of freedom
AIC: 69.02
Number of Fisher Scoring iterations: 6

Rostrum: The coefficient for Rostrum is -0.09456. It is statistically significant ($p < 0.001$), indicating that Rostrum has a significant effect on the log-odds of the response variable.
Model Fit:

Null Deviance: 123.28 (baseline deviance)
Residual Deviance: 65.02 (model deviance)
AIC: 69.02 (model fit statistic)

Model 7
Summary Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.8723	-0.5825	0.1522	0.4403	1.9275

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	56.46070	11.61796	4.860	1.18e-06***
Height	-0.07223	0.01489	-4.851	1.23e-06***

Null deviance: 123.279 on 88 degrees of freedom
Residual deviance: 67.579 on 87 degrees of freedom
AIC: 71.579
Number of Fisher Scoring iterations: 6

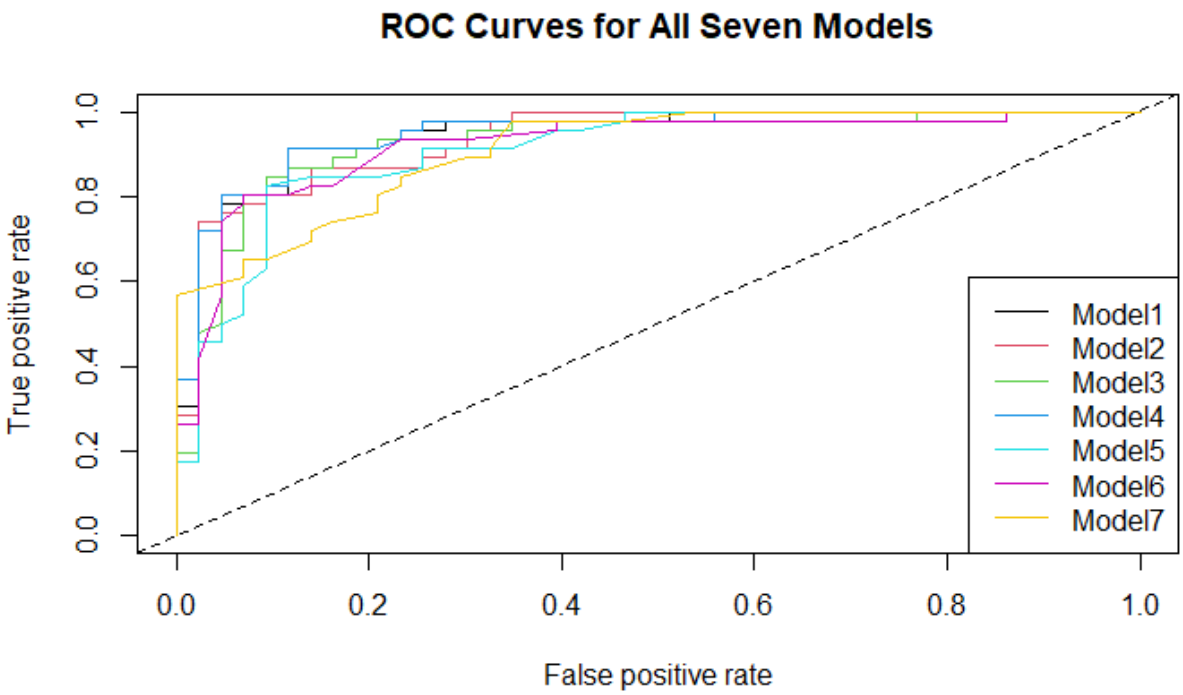
Intercept: The intercept is 56.46070.
Height: The coefficient for Height is -0.07223. It is statistically significant ($p < 0.001$), indicating that Height has a significant effect on the log-odds of the response variable.
Model Fit:

Null Deviance: 123.279 (baseline deviance)
Residual Deviance: 67.579 (model deviance)
AIC: 71.579 (model fit statistic)

4. Comparing Error Rates, AIC and AUC of the Model

Models	Error_rates
Model1	0.09835572
Model2	0.10838096
Model3	0.10904786
Model4	0.09590377
Model5	0.12911468
Model6	0.11498569
Model7	0.13555802

Models	AIC
Model1	58.62486
Model2	61.48884
Model3	67.31891
Model4	56.96254
Model5	73.39014
Model6	69.02009
Model7	71.57911



Models	AUC
Model 1	0.946916077
Model 2	0.936299292
Model 3	0.925935288
Model 4	0.94817998
Model 5	0.909251769
Model 6	0.919868554
Model 7	0.90950455

Model4 has the lowest AIC value (56.96254), indicating that it has the best balance of model fit and complexity according to the AIC criterion.
However, considering the error rates. A lower error rate indicates better predictive performance. Among the models with relatively low AIC values, Model4 also has one of the lowest error rates (0.09590377).
Model 4 has the highest Area Under Curve (AUC) value of 0.9482, indicating that it performs the best among all the models in distinguishing between the classes "multiples" and "subterraneus." This means that Model 4 is highly effective at correctly classifying instances of both classes and has the strongest discriminatory power.
Model 1 is the second-best performer with an AUC of 0.9469, also demonstrating excellent classification ability. It comes very close to the performance of Model 4 but falls just slightly behind.

Therefore, based on the AIC and error rates and AUC, Model4 seems to be the preferred choice as it has a good balance between model fit (as indicated by the AIC) and predictive performance (as indicated by the error rate).

In summary, Model4 outperforms the others in terms of predictive accuracy, as indicated by its lowest error rate. Model7, on the other hand, shows the least accuracy. The remaining models display varying degrees of performance, with Model1 also showing relatively strong results.

5. Predictions for known Species

Since Model 1 and Model 4 had low error rate and AIC we applied confusion matrix in both the models.

The following table shows the test of **Model 1** with the known species:

Species	Multiplex	Subterraneus
Multiplex	38	5
Subterraneus	4	42

The following table shows the test of **Model 2** with the known species:

Species	Multiplex	Subterraneus
Multiplex	37	6
Subterraneus	9	37

The following table shows the test of **Model 3** with the known species:

Species	Multiplex	Subterraneus
Multiplex	38	5
Subterraneus	7	39

The following table shows the test of **Model 4** with the known species:

Species	Multiplex	Subterraneus
Multiplex	38	5
Subterraneus	4	42

The following table shows the test of **Model 5** with the known species:

Species	Multiplex	Subterraneus
Multiplex	34	9
Subterraneus	7	39

The following table shows the test of **Model 6** with the known species:

Species	Multiplex	Subterraneus
Multiplex	37	6
Subterraneus	8	38

The following table shows the test of **Model 7** with the known species:

Species	Multiplex	Subterraneus
Multiplex	34	9
Subterraneus	9	37

All the models were tested for identifying two species: Multiplex and Subterraneus. Remarkably, they performed the same. Interestingly model 1 and model 4 both correctly identified 38 instances of Multiplex and 42 of Subterraneus. However, both mistakenly classified 5 Subterraneus as Multiplex and missed 4 Multiplex, labeling them as Subterraneus.

5. Predictions for Unknown Species

The table below demonstrates the prediction of unknown species **(Model 1)**

Species	Multiplex	Subterraneus	Total
Unknown	69	130	199

The table below demonstrates the prediction of unknown species **(Model 2)**

Species	Multiplex	Subterraneus	Total
Unknown	76	123	199

The table below demonstrates the prediction of unknown species **(Model 3)**

Species	Multiplex	Subterraneus	Total
Unknown	73	126	199

The table below demonstrates the prediction of unknown species **(Model 4)**

Species	Multiplex	Subterraneus	Total
Unknown	77	122	199

The table below demonstrates the prediction of unknown species **(Model 5)**

Species	Multiplex	Subterraneus	Total
Unknown	84	115	199

The table below demonstrates the prediction of unknown species **(Model 6)**

Species	Multiplex	Subterraneus	Total
Unknown	97	102	199

The table below demonstrates the prediction of unknown species **(Model 7)**

Species	Multiplex	Subterraneus	Total
Unknown	93	106	199

The models were employed to predict whether unknown species belonged to either the "Multiplex" or "Subterraneous" categories.

When all three variables (length, height, and rostrum) are available, Model 1 demonstrated low error rates and balanced predictions for both Multiplex and Subterraneus species. Therefore, Model 1 is a suitable choice for predicting Unknown species classification under these conditions. However, in cases where one or more variables are missing, it is crucial to select models that can utilize the available data. For instance, if only the length variable is available, Model 5, which includes only the length variable, should be chosen for classification.

Usefulness of Predictions:

- **Practical Application:** The accurate classification of species is essential for various scientific and ecological purposes, such as understanding population dynamics, habitat management, and conservation efforts. Accurate predictions help researchers and conservationists make informed decisions.
- **Model Reliability:** Model 4, in particular, with its lowest AIC value and high AUC, is considered a reliable tool for species classification. However, when height and rostrum variables are available it is a good option.
- **Data Quality Improvement:** The models have been trained on cleaned and corrected data, which helps improve the quality of the dataset by correcting errors and outliers. This enhanced dataset can be valuable for future research and analyses.

- **Research Insights:** Accurate species classification can provide valuable insights into the behavior, ecology, and distribution of Microtus species, contributing to a deeper understanding of these rodents.
- **Conservation Impact:** In cases where certain Microtus species are endangered or threatened, accurate classification can aid in designing targeted conservation strategies to protect these species and their habitats.
- **Efficiency:** Using models for classification is often faster and more efficient than manual identification, especially when dealing with many specimens.

In conclusion, the predictions made by Models is useful for scientific research, ecological studies, and conservation efforts. They offer a reliable and efficient means of identifying Microtus species, contributing to better-informed decision-making and a deeper understanding of these rodents' biology and ecology.

Reference

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An Introduction to Statistical Learning. Springer.

Airoldi, J.-P., B. Flury, M. Salvioni (1996) “Discrimination between two species of Microtus using both classified and unclassified observations” Journal of Theoretical Biology 177:247-262

https://www.youtube.com/watch?v=X12BoYpIjyw

https://www.dataquest.io/blog/control-structures-in-r-using-loops-and-if-else-statements/